H Govardhan Reddy
192425096
MLA0205
Fundamentals of machine learning.

CIA-1

② Given data: joint distribution P(x,y) of load x (low, medium, High) and fault y (No, yes), model Q = (0.5, 0.3, 0.2)

load x	y = No	y = Yes	P(x)
low	0.30	0.02	0.32
medium	0.28	0.10	0.38
High	0.10	0.20	0.30
P(y)	0.68	0.32	1.00

30/8

Numerical solution:-

1) Margins: P(x) = (0.32, 0.38, 0.30); P(y) = (0.68, 0.32).
Row and column sums of the joint table.
2) H(x) = $-\Sigma P \log_2 P$ = 0.5260 + 0.5304 + 0.5211 = 1.5776 bits.
H(y) = 0.3784 + 0.5260 = 0.9044 bits.
3) H(x,y) = 0.5211 + 0.1129 + 0.5142 + 0.3322 + 0.3322 + 0.4644
= 2.2770 bits.
4) H(y|x) = H(x,y) − H(x) = 2.2770 − 1.5776 = 0.6994 bits.
5) I(x,y) = H(y) − H(y|x) = 0.9044 − 0.6994 = 0.2050 bits (= H(x) + H(y)
− H(x,y)
6) Crossentropy H(P_x, Q) = $-\Sigma P_x \log_2 Q$) = 0.32 (1.0000) +
0.38 (1.7370) + 0.30 (2.3219) = 1.6766 bits.
7) KL divergence D_KL (P_x | Q) = H(P_x | Q) − H(x) = 1.6799
− 1.5776 = 0.0990 bits
8) P(y = yes | x = High) = 0.20/0.30 = 0.6667 (Low: 0.0625; Medium: 0.2632) Fault probability rises sharply with load with the overall P(yes) = 0.32.

Interpretation and conclusion: The Mutual informati
0.205 bits is positive, so load carries useful bu
moderate information about faults (high load → 66.7 %
Probability versus 6.25 % at low load). Model Q has D
≈0.099 bits. It over-weights low load (0.5 vs 0.32) and
under-weight High load (0.2 vs 0.30) re-estimated from
the observed frequencies.

1) Given data P(C) = 0.03; sensitivity P(+|C) = 0.90; specificity
P(−|¬C) = 0.94; loss of not treating a diseased patient
λ(no|C) = 50; loss of treating a healthy patient λ(treat|¬C)
= 2; correct decisions cost 0.

Numerical solution:-

1) Priors and likehoods P(¬C) = 0.97; P(+|C) = 0.90; P(+|¬C)
= 1 − 0.94 = 0.06; P(−|C) = 0.10 P(−|¬C) = 0.94

2) Evidence P(+) = 0.90 × 0.03 + 0.06 × 0.97 = 0.0270 + 0.0582
= 0.0852; P(−) = 1 − 0.0852 = 0.94148 Total Probability combines.

3) Posteriors: P(C|+) = 0.0270/0.0852 = 0.3169; P(¬C|+)
= 0.6831; P(C|−) = (0.10 × 0.03)/0.9148 = 0.0033; P(¬C|−) =
0.9967.

4) Conditional risks: R(treat|+) = 2 × 0.6831 = 1.3662; R(no
treat|+) = 50 × 0.3169 = 15.845; R(treat|−) = 2 × 0.9967 =
1.9934; R(not treat−) = 50 × 0.0033 = 0.1640.

5) Minimum-risk decisions positive → treat (1.3662 < 15.845)
negative → do not treat (0.1640 < 1.9934) P(C|x) > 2/(2+50) =
0.0385. P(C|+) = 0.3169 exceeds.

6) Bayes risk: R = 0.0852 × 1.3662 + 0.9148 × 0.1640 = 0.1164 +
0.1500 = 0.2664 check: 2 × P(¬C, +) + 50 × P(C, −) = 2 × 0.0582
+ 50 × 0.0030 = 0.2664.

7) second independent positive test: Posterior odds =
(0.3169/0.6831) × (0.90/0.06) = 0.4639 × 15 = 6.959; P(C|++)
= 6.959/7.959 = 0.8744

imum error (0-1 loss) rule: decide c only if $P(c|x) > 0.5$ — Govardhan Reddy

ice $P(c|+) = 0.3169 < 0.5$ the rule never treats, its expected
ost under the real loss matrix $= 50 \times 0.033 = 1.5$ units per Patient

Interpretation and conclusion:-

A Positive result given only a 31.7% chance of disease, ye
treatment is justified because a missed costs 25 times a
false alarm. The minimum-risk rule (treat on +, no treat on -
gives a Bayes risk of 0.2664 and is far more suitable
than the minimum error $P(c)$ to 87.44%, supporting confirma
tory testing.

③ (i) Given data samples $x = \{48, 52, 55, 47, 53, 51\}$; $n = 6$.

Numerical Solution:-

1. ML Mean: $\mu = (48+52+55+47+53+51)/6 = 306/6 = 51.0°c$
The ML estimate of the Gaussian mean is the sample mean.

2 - squared deviations $(-3)^2 + 1^2 + 4^2 + (-4)^2 + 2^2 + 0^2 = 9+1+16+16+4+0 = 46$. ML variance $\sigma^2 = 46/6 = 7.667$. $\sigma = 2.769°c$

3: unbaised variance $s^2 = 46/5 = 9.20$; $s = 3.033°c$
Dividing by $n-1$ corrects the bias because $E[\sigma^2_{ML}] = ((n-1)/n)\sigma^2 = (5/6)\sigma^2$ the ML estimates $\sigma^2 = 46/6 = 7.667$;

4. $P(x > 55)$: $z = (55-51)/(2.769) = \pm 1.083 = 0.0743$
Table values $\phi(1.44) = 0.9251$ and $\phi(1.45) = 0.9265$ given
$0.07 - 0.075$ (accept 0.074 ± 0.001) with $s = 3.033$ the tail
becomes 0.0936.

5) $P(48 < x < 54)$: $z = \pm(3/2.769) = \pm 1.083$; $P = 2\phi(1.083) - 1 = 0.72$
$1.4 \approx 0.72$ Accept 0.72 ± 0.01 using $\phi(1.08) = 0.8599$.

Interpretation and conclusion:-

The ML estimates are $\mu = 51.0°c$ and $\sigma^2 = 7.667°c^2$ the
unbaised variance is larger (9.20). for small smaples the ML
variance understimates the true variance and therefore
understimates Probabilites (0.0743) aganist 0.0936 for
$P(x > 55)$, so the unbaised estimator is Preferred when
n is small.

③ (ii) Given data (x,y) = (10,25),(20,38),(30,52),(40,61),(50,

ridge λ=400 on centred data.

Numerical Solution:

1) Means $\bar{x}$ = 30; $\bar{y}$ = (25+38+52+61+79)/5 = 51.3; S_{xx} = Σ(

= 400+100+0+100+400 = 1,000; $S_{xy} = \Sigma(x-\bar{x})(y-\bar{y})$ = 52

+130+0+100+560 = 1,310

$w_1 = S_{xy}/S_{xx}$ and $w_0 = \bar{y} - w_1\bar{x}$.

2) w_1 = 1,310/1,000 = 1.31 ms per %; w_0 = 51 − 1.31×30 = 11.7 ms

fitted model: $\hat{y}$ = 11.7 + 1.31x.

3) fitted values: 24.8, 37.9, 51.0, 64.1, 77.2 residuals 0.2

0.1, 1.0, −3.1, 1.8; SSE = 0.04+0.01+1.00+9.61+3.24 = 13.

4) SST = 676+169+1+100+784 = 1,730; R^2 = 1 − 13.90/1,730 =

0.9920

The line explains 99.2% the variation in response time

5) Prediction at x = 60; $\hat{y}$ = 11.7 + 1.31×60 = 90.3 ms

x=60 lies outside the training range 10-50, so this

is explanation.

Integretation and conclusion:-

With only five clean perfectly linear points λ=500 over regularises and under-fits (R^2 drops from 0.992 to 0. baising the 60% prediction down to 77.2m The ordina least-squares model (90.3 ms) is more reliable here ; small λ chosen by cross−validation treated cautiousl because it is extrapolated.